

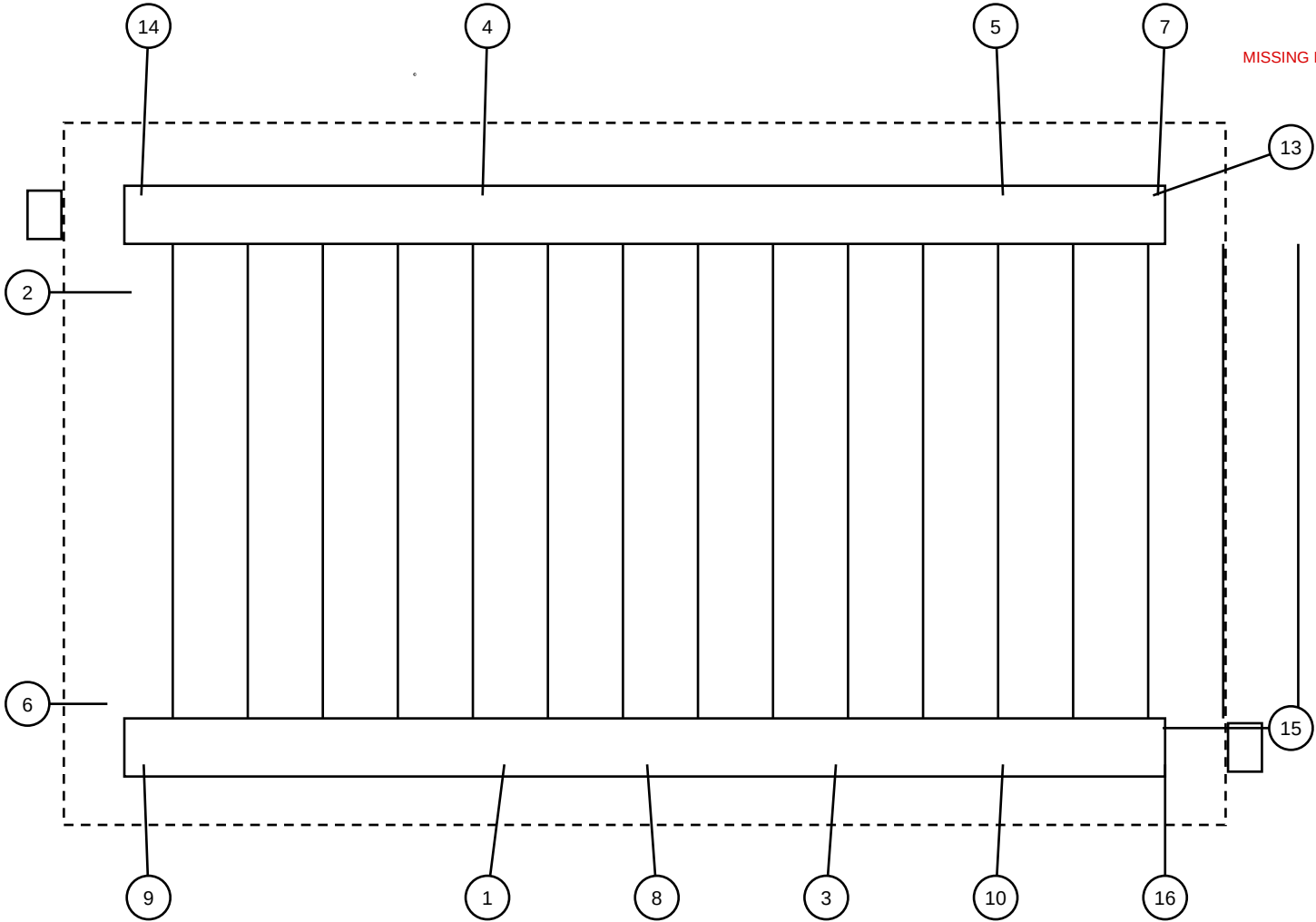
BOM CHECK (automated - verify before acting)

red = on drawing, missing in BOM

orange = qty / spec / revision mismatch

yellow = confirm match / extra / duplicate

blue = suggestion (implied item, rule-based)



NOTES:

1. ALL DIMENSIONS ARE IN MM.
2. ALL WELDING AS PER IBR / ASME SEC I.
3. INSULATE WITH 100 THK MINERAL WOOL AND CLAD WITH 0.6 THK ALUMINIUM SHEET. <- BOM LACKS: Cladding

| Item | Description                       | Material          | Qty | Part No  | Remarks |
|------|-----------------------------------|-------------------|-----|----------|---------|
| 1    | INLET HEADER 219.1 OD X 8.0 THK   | SA 106 GR.B       | 1   | EC-01-01 |         |
| 2    | OUTLET HEADER 219.1 OD X 8.0 THK  | SA 106 GR.B       | 1   | EC-01-02 |         |
| 3    | ECONOMISER TUBE 38.1 OD X 3.6 THK | SA 210 GR.A1      | 140 | EC-01-03 |         |
| 4    | RETURN BEND 38.1 OD               | SA 210 GR.A1      | 24  | EC-01-04 |         |
| 5    | TUBE SUPPORT PLATE 12 THK         | IS 2062 GR.B      | 11  | EC-01-05 |         |
| 6    | INLET FLANGE 200 NB CL.150        | SA 105            | 1   | EC-01-06 |         |
| 7    | OUTLET FLANGE 200 NB CL.150       | SA 105            | 1   | EC-01-07 |         |
| 8    | DRAIN VALVE 25 NB                 | SA 105            | 2   | EC-01-08 |         |
| 9    | VENT VALVE 15 NB                  | SA 105            | 1   | EC-01-09 |         |
| 10   | THERMOWELL 1/2 NPT                | SS 316            | 2   | EC-01-10 |         |
| 11   | GASKET 200 NB CL.150              | SS 304 + GRAPHITE | 2   | EC-01-11 |         |
| 12   | STUD BOLT M20 X 90 WITH 2 NUTS    | SA 193 B7         | 16  | EC-01-12 |         |
| 13   | LIFTING LUG 16 THK                | IS 2062 GR.B      | 4   | EC-01-13 |         |
| 14   | CASING PLATE 3 THK                | IS 2062           | 18  | EC-01-14 |         |
| 15   | EXPANSION JOINT 600 X 600         | FABRIC            | 1   | EC-01-15 |         |
| 16   | INSPECTION DOOR 300 X 300         | IS 2062           | 2   | EC-01-16 |         |

ECONOMISER ASSEMBLY

DRG NO : EC-01-GA-001

DRG REV : B

SCALE : NTS    SIZE : A3

SAMPLE (SYNTHETIC) DRAWING

# BOM CHECK SUMMARY

Drawing : sample\_01\_drawing.pdf (Rev B)  
BOM : sample\_01\_bom.xlsx (Rev B)  
Input : vector PDF  
RUN RELIABILITY : HIGH

Drawing items checked : 16  
BOM lines read : 18  
BOM completeness : 93.8% (drawing items that have a BOM line)

Findings: duplicate 1, implied 1, omission 1, spec 1

[RED ] Item 12 'STUD BOLT M20 X 90 WITH 2 NUTS' (qty 16) is on the drawing but has no BOM line  
[ORANGE] Item 6: specification mismatch - dimension: drawing ['200'] vs BOM ['250']  
[YELLOW] BOM row 23 'Outlet header 219.1 OD x 8 thk' repeats item 2 (already covered by BOM row 7)  
[BLUE ] Suggestion (Cladding): the drawing notes imply it, but no matching line was found in the BOM

Automated check - not a substitute for engineering review. Blue items are rule-based suggestions.